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Assessing the value of artificial intelligence (AI) in governmental public procurement

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Abstract

Purpose – The main purpose of this study is to enhance knowledge regarding the early stages of planning for and adopting artificial intelligence (AI) in governmental public procurement. While there are numerous studies on AI and procurement in private companies, there is limited information on AI and public procurement.

Design/methodology/approach – The empirical data consists of information obtained from 18 semi-structured interviews with procurement managers and individuals involved in the development of procurement at governmental agencies. Additionally, a workshop was conducted with the respondents to discuss and validate the study's findings.

Findings – Findings indicate a generally low level of AI maturity in previous research and within the investigated governmental agencies. The perceived benefits of AI primarily revolve around improved operational capabilities, potential for certain process efficiencies and the ability to enhance monitoring through AI. Various challenges related to organizational, process, technological and data management were highlighted. Findings also indicate that perceived benefits and value created by AI can be viewed from a short-term perspective to a long-term perspective.

Social implications – The study provides insights into societal values that can be achieved using AI in public procurement.

Originality/value – This study provides a new perspective on AI in public procurement by focusing on governmental agencies. It explores the perceived benefits, interests and challenges associated with AI implementation in public procurement. Furthermore, this study discusses the potential outcomes of incorporating AI in public procurement and the impact it may have on the values created by the public service, both short- and long term.

Keywords Public sector, Procurement, Digital, Information and communications technology, Value

Paper type Research paper

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Introduction

Artificial intelligence (AI) has been identified as having significant potential in transforming procurement operations and enhancing value creation in private business procurement activities (Constant *et al.*, 2022; Cui *et al.*, 2022; Allal-Chérif *et al.*, 2021; Rejeb *et al.*, 2018). Kosmol *et al.* (2019) suggests that digitalization, including AI, can provide additional value to procurement. Similarly, Rejeb *et al.* (2018) state in their technology review that "the combined usage of robotics along with artificial intelligence (AI) and machine learning (ML) clears the way for significant contributions to the field of procurement" (p.79).

Despite the increasing importance of procurement for governments and countries, there is a limited number of studies on AI and public procurement (Sanchez-Graells, 2024a; Siciliani *et al.*, 2023; Abrahamsson and Behr, 2022; Nowicki, 2020). Numerous studies have focused on AI and private sector procurement (Meyer and Henke, 2023; Guida *et al.*, 2023; Constant *et al.*, 2022; Cui *et al.*, 2022; Allal-Chérif *et al.*, 2021; Kosmol *et al.*, 2019; Rejeb *et al.*, 2018), but few have specifically examined AI and its impact on public procurement. The use of AI in public administration including public procurement is riddled with ethical tensions of fairness, transparency, privacy and more, and a large knowledge gap remains in understanding AI tensions as they relate to public value creation (Madan and Ashok, 2023). Additionally, studies on AI and procurement often lack detailed qualitative empirical data (Guida *et al.*, 2023; Allal-Chérif *et al.*, 2021). Therefore, there is a need for more knowledge and qualitative empirical data to advance the field of AI in public procurement.

The implementation of AI in public procurement presents numerous opportunities and benefits, however, like public management in general, it also comes with apparent risks, obstacles and inertia (Wirtz *et al.*, 2021; Wirtz and Muller, 2019; Wirtz *et al.*, 2019). The adoption of AI in private business procurement operations is undergoing significant changes (Allal-Chérif *et al.*, 2021), and its potential for public procurement has also been recognized (Nowicki, 2020). Nevertheless, research on AI in public procurement is still in its early stages, and more knowledge is needed about initial phases of assessing the value, planning for and adopting AI in public procurement (Nowicki, 2020). Therefore, the aim of this paper is to develop knowledge regarding the perceived benefits of AI in public procurement and the challenges associated with its implementation, as well as how AI can contribute to the creation of value. In response to the call for in-depth qualitative studies (Wirtz *et al.*, 2021), this research utilizes a series of in-depth interviews with procurement professionals at ten governmental agencies. Building upon research on AI in public management, public procurement studies and recent research on AI in procurement, the study is guided by two research questions:

RQ1. What are the initially perceived benefits of AI in public procurement and how do they relate to the general values created by public procurement?

RQ2. What are the challenges when implementing and trying to achieve these benefits?

The purpose of this article is to address gaps in public management research on AI and in public and general procurement research, where there is a need for empirical studies to enhance understanding of the reasons behind AI adoption, implementation processes, challenges and effects (e.g. Guida *et al.*, 2023). In this study AI is viewed as a multidisciplinary technology with the capability to integrate cognition, machine learning, emotion recognition, human-computer interaction, data storage and decision-making (Zhang and Lu, 2021; Lu, 2019). The study is based on qualitative interviews with representatives from 10 governmental agencies' procurement functions, along with a workshop involving participants from the interview study. The article is structured as follows: first a literature review is presented where we report on previous studies within the field. This is followed by a presentation of theoretical framework

used for the analysis. Methodology is thereafter presented. Following is empirical results, analysis and discussion of the main findings. Conclusions are then provided, along with theoretical and managerial implications. The paper concludes with limitations and suggestions for future studies.

Literature review

Artificial intelligence and public procurement

Previous studies that have investigated and discussed AI and public procurement have done so mainly from a law and information system perspective (Sanchez-Graells, 2024a; Sanchez-Graells, 2024b; Siciliani *et al.*, 2023; Abrahamsson and Behr, 2022; Nowicki, 2020). Studies from a business administration perspective or from a public procurement perspective are rare. Also empirical studies are rare, this probably due to that use of AI in public procurement is still in an early stage (Nowicki, 2020). In existing studies, it is argued that new technologies such as AI will undoubtedly enrich and improve the public procurement system (Nowicki, 2020). Nowicki (2020) further argues that machine learning and AI offer great opportunities especially in planning and preparation of the awarding procedures concerning individual contracts, as well as managing the public procurement contract. A suitable computer program for example could collect data related to the subject matter of the procurement, analyze the performance of similar contracts and then compare them (Nowicki, 2020). In the preparation phase authorities should get to know the best market, to check available solutions, to assess their cost throughout the entire life cycle and to estimate risks occurring in a given market, this should be possible using AI technologies (Nowicki, 2020). According to Nowicki (2020) the program could also on the basis of appropriate data, manage the execution of public procurement contracts such as monitoring the progress of construction work and informing the contracting authority about any possible delays coming from changes in weather forecasts or the availability of necessary materials.

There are several opportunities brought up by previous studies on AI technologies and public procurement, but also challenges such as ethical, legal and safety concerns (Sanchez-Graells, 2024b; Abrahamsson and Behr, 2022; Nowicki, 2020). Using new technologies such as AI technologies means that we are processing huge amounts of data and there can be doubt about their security, this in relation to the right of data protection (Nowicki, 2020). Nowicki (2020) also brings up the importance of the right amount and quality of data as a condition for obtaining appropriate results when using machine learning and AI in public procurement. To gain the right amount and quality of data can be challenging, and how do we make sure it is the right amount and quality. According to Nowicki (2020) it seems a difficult task to ensure exponential growth in the field of AI, as well as to create a legal regulation that is flexible enough to adopt to it without fear that it will soon become outdated, while ensuring adequate security and legal clarity.

Artificial intelligence and private procurement

Since studies on AI and public procurement are few, we might learn from studies on AI and private procurement. Even though there are different contexts, conditions and requirements, such as greater demand on transparency in public procurement and also other ethical and legal conditions and requirements (Nowicki, 2020; Sanchez-Graells, 2024a; Sanchez-Graells, 2024b), it is relevant also to consider, discuss and relate to studies on AI and procurement in the private sector.

Research on the implementation of AI technologies in private procurement is also in its early stage much because procurement is still lagging in AI adoption as compared to other business areas (Spreitzenbarth *et al.*, 2024; Spreitzenbarth *et al.*, 2021). And the possible reasons of lagging are challenges, barriers and uncertainties in adoption or implementation.

The sporadic literature on AI and procurement highlights the use cases and potential benefits of implementation of AI systems and tools in procurement processes that claims to add value to it (Cui *et al.*, 2022). Many studies have cited its potential to transform various procurement processes like cost estimation, auctions, risk management, etc. and as a result, digitally transformed procurement organizations, utilizing AI and data analytics are efficient in their operations (Niyazbekova *et al.*, 2021). AI can be a tool in bidding, supplier identification and supplier evaluation (Jahani *et al.*, 2021). A study highlighted that AI along with cognitive analytics can unveil the most potential within planning-to-strategy and source-to-contract layers of procurement process (Buchholz and Grabbe, 2022). Therefore, negotiations being the crucial part of procurement processes in source-to-contract can benefit from AI as AI simulations can support the design-based complex mechanisms to achieve the best outcomes and quotations (Schulze-Horn *et al.*, 2020). Furthermore, AI can also aid procurement in optimizing the forecasting capabilities using machine learning techniques (Kiefer *et al.*, 2019). Challenges to AI implementation raised in previous studies are availability and quality of data, cultural barriers such as lack of internal analytical skills and digital maturity, also uncertainty about the real applications of AI and low awareness of the technology (Guida *et al.*, 2023).

Theoretical framework

To analyze the two research questions, we primarily rely on research and conceptualizations in three areas: recent procurement research focused on AI technologies (Allal-Chérif *et al.*, 2021; Cui *et al.*, 2022), information management research focused on AI implementation challenges in business organizations (Merhi, 2023; Enholm *et al.*, 2021; Kaplan and Haenlein, 2020;) and public procurement research focused on value creation (Malacina *et al.*, 2022).

Benefits and affordances of artificial intelligence in public procurement

Our first research question examines the perceived *benefits* of AI in public procurement. In their extensive literature review of AI research in public management, Wirtz *et al.* (2021) noted the lack of an existing classification for AI benefits. They concluded that previous studies have not provided a comprehensive understanding of benefits of AI. Therefore, we approach our first research question inductively, relying on the interviewees' perceptions of AI's potential in governmental procurement. However, we can also compare these perceptions with the benefit categories extracted by Wirtz *et al.* (2021) for AI in public management, which include *broader accessibility/availability, competitive advantage, cost effectiveness, ecological impact, information processing, labor-saving, qualitative result effectiveness, quantitative result effectiveness, time-saving and transparency.*

The focus here on perceived benefits leads directly to theoretical ideas on (technological) affordances. When actions related to perceived AI benefits are implemented the potentials of technology and technical objects - *affordances* - are actualized (Markus and Silver, 2008). Building on Strong *et al.* (2014), Keller *et al.* (2019) concluded that "since affordances are possibilities for actions, actors must take these actions to achieve the relevant outcomes" (Strong *et al.*, 2014). In this study, interviewees discuss both the generally perceived values but also various actions to realize these benefits/technological affordances. Various implementation challenges are perceived, and ideas are presented on how they can be managed. These actions can then finally result in both first- and second order effects and the outcomes can be seen in various changes in public value.

Effects on public value

The first research focus also draws attention to the *value* of AI of public procurement when perceived benefits and affordances are realized in public services. We approach this value from procurement research and AI focused IT management research. We build on Malacina *et al.* (2022) and how they extract and distinguish between six key public purchasing and supply chain (PSM) practices; four internal (vertically aligned PSM practices, enabling PSM practices, within PSM practices, cross-functional PSM practices) and two external (relational PSM practices, non-relational practices). Second, in a matrix these six can be seen to be linked to different degrees to six general value components realized by public procurement: *sustainability, market development and performance, innovation promotion, better operative capabilities, public procurement process effectiveness* and *quality and availability of product/service*. The perceived value outcomes from AI in procurement is in IT management research described as *first-order and second-order effects of AI implementation* (Enholm *et al.*, 2021). First-order effects are the effects on focal processes (here: procurement) while the second-order effects of AI implementation are the effects on the overall organization. To the first category belong process efficiency, insight generation and business process transformation. The second order effects concern overall operational performance, financial performance, market-based performance, sustainability performance and also unintended consequences.

Artificial intelligence implementation challenges

The second research question asks what the key *challenges* are when *implementing* AI technologies in public procurement. Here, two broad AI research overviews (Merhi, 2023; Enholm *et al.*, 2021;) have listed enablers/inhibitors and critical factors associated with AI implementation processes in general. The identified factors largely overlap. Merhi (2023) groups AI implementation challenges into four different types:

(1) organization (top management support, ambiguous strategic vision, organizational culture, organizational structure, lack of visibility on benefits);

(2) process (project champion, resistance, lack of technical expertise, ethics, responsibility and accountability);

(3) technology (integration complexity, low data quality, insufficient quantity of data, it infrastructure, security and confidentiality, data governance issues, scalable and flexible system); and

(4) environment (selection of vendors, high cost of AI).

The research question focuses on the public procurement processes that are most relevant for AI. The Swedish Competition Authority (2016) divides the process into three main phases: preparation phase (demand and need analysis, supply market research and analysis, strategy development), execution phase (tendering and supplier selection) and contract management phase (expediting and evaluation, follow-up and evaluation). This division is used to frame the empirical results.

Summing up, in a technological context, perceived AI benefits and affordances are the possibility for goal-oriented actions in public procurement. As perceived AI benefits are implemented and affordances are actualized, various implementation challenges are perceived and managed. These actions can result in both first- and second order effects and the outcomes can be seen in various changes in public value. These aspects and concepts are in focus of our analysis of AI in public procurement.

Methodology

Since literature on AI and public procurement is in its infancy, this study has taken an exploratory qualitative approach enabling us to explore this new phenomenon in context (Yin, 2018). For data collection, semi-structured interviews were used to gain a deeper understanding of the reasoning behind the public sector, public procurement processes and AI solutions (Yin, 2018). Altogether 18 semi-structured interviews were conducted with procurement managers and individuals involved in procurement development at 10 Swedish government agencies, as well as one business support organization responsible for managing procurement for agencies. In a preliminary study, four individuals were interviewed, and in the main study, 15 individuals. One individual was interviewed in both the preliminary and main study due to his/her dual role. Additionally, a workshop was conducted with respondents to discuss and confirm the study's results.

Choice of context

In Sweden, public procurement amounts to over 80 billion euros annually, which is nearly one-fifth of the country's GDP (20%). In comparison, the average amount spent on public procurement in OECD countries is 12% of GDP. Consequently, public procurement in Sweden has a significant impact on the country's economic development and has the potential to contribute to societal value through the use of technologies like AI. In addition, the Swedish public sector is an interesting case due the challenge the public sector is facing. It is increasingly squeezed between an ageing population and a shrinking tax base, a situation that implies need for innovation in public procurement processes (Knutsson and Thomasson, 2014).

Preliminary study to identify suitable government agencies

To identify the most suitable agencies to include in the study, a preliminary study was conducted. This study involved individuals with insight and extensive knowledge of government agencies' procurement work. Through the preliminary study, relevant and interesting agencies were identified for inclusion in the main study.

Semi-structured interviews with public procurement professionals

Semi-structured interviews were conducted with individuals working in procurement at the selected government agencies from June 2022 to December 2022. In total, 15 individuals from 9 government agencies and a business support organization responsible for managing procurement for agencies were interviewed in the main study. See Table 1 below for roles interviewed and areas of responsibility for organizations included in the study. Interviews took place via Zoom or Teams and lasted between 50 and 70 min.

The interview questions were guided by the research questions and the analytical framework following a standardized format but also allowing for flexibility, for new questions to be brought up during the interview (Saunders *et al.*, 2023; Seale *et al.*, 2004).

Data analysis

The analytical process implied a thematic qualitive analysis (Saunders *et al.*, 2023; Braun and Clarke, 2006). The researchers began by familiarizing themselves with the data. All interviews were recorded and then transcribed with the support of the software Sonix. In line with Braun and Clarke (2006) the researchers started to become familiar with the data through reading and re-reading the interview protocols, also noting down initial ideas. The thematic analysis had more of an abductive approach in which the theoretical framework played a guiding, but not determining role when coding the data (Thompson, 2022).

Table 1. Areas of responsibility for organizations included in the study and roles interviewed

Organization	Organization/agency area of responsibility	Role	Type of interview	Date
Organization 1	Conducts and coordinates Sweden's environmental work	Procurement manager	Zoom	June 27, 2022
Organization 2	Manages civil registration of private individuals and collects taxes	Procurement manager	Teams	Oct 27, 2022
Organization 3	An operating support for the public sector, owned by Sweden's municipalities and regions. Offers public sector contracts and services in strategic supply	Manager procurement and business law	Teams	Nov 9, 2022
Organization 4	Preserves and develops state-owned cultural and historical properties	Procurement manager	Teams	Nov 17, 2022
Organization 5	Responsible for the public employment service and the implementation of labor market policies	Procurement manager	Teams	Nov 18, 2022
Organization 6	Procures, develops and delivers technology, services and materiel to the Swedish armed forces	Manager processes procurement	Teams	Nov 24, 2022
Organization 6	Procures, develops and delivers technology, services and materiel to the Swedish armed forces	Expert procurement systems consultant	Teams	Nov 24, 2022
Organization 6	Procures, develops and delivers technology, services and materiel to the Swedish armed forces	Purchaser	Teams	Nov 24, 2022
Organization 7	Responsibility to see to that work environment and working hours are followed by companies and organizations	Deputy procurement manager	Teams	Nov 25, 2022
Organization 8	Responsible for long-term planning of infrastructure	Innovation strategist	Teams	Dec 2, 2022
Organization 8	Responsible for long-term planning of infrastructure	Project manager procurement development	Teams	Dec 2, 2022
Organization 8	Responsible for long-term planning of infrastructure	Procurement and logistics strategist	Teams	Dec 2, 2022
Organization 8	Responsible for long-term planning of infrastructure	Procurement controller	Teams	Dec 2, 2022
Organization 9	Responsible for helping society prepare for major accidents, crises and the consequences of war	Purchaser	Teams	Dec 9, 2022
Organization 10	Responsible for preventing crime, monitoring public order and safety, conducting reconnaissance and carry out criminal investigations	Head of section procurement	Teams	Dec 15, 2022

Source: Authors' own creation

Coding was conducted by going through the interview protocols one by one labelling each unit of data, which in our case consisted of sentences or a section of text, with a code that summarized that extract's meaning (Saunders *et al.*, 2023). After coding the data, the researchers organized

codes by drawing them together into themes (Saunders *et al.*, 2023). Some codes formed main themes whereas others formed sub-themes (Braun and Clarke, 2006). When coding the data the researchers were guided by the theoretical framework consisting of benefits of AI in public procurement, effects on public value and AI implementation challenges, but also empirical data falling outside the scope of the theoretical framework was included in the data analysis and coded.

Workshop

After completing the interview study and analysis, the results were discussed in a hybrid workshop on November 17, 2023. Twelve individuals participated, all of whom were interviewed in the main study. Four participants joined online, and eight were present in person, representing eight of the ten organizations. The workshop, led by two researchers, lasted for 2 h. The results were presented and discussed, and conclusions were confirmed.

Empirical findings

Great interest in artificial intelligence and public procurement but few artificial intelligence applications currently in use

There is a large interest in using AI applications in public procurement, as evidenced by the preliminary and main study. Respondents expressed a common belief that AI applications have the potential to add value to the public procurement process. However, out of the 10 organizations included in the main study, only two currently utilize AI applications in procurement. Nevertheless, the other eight organizations were informed about AI and acknowledged its potential use. One of these organizations had recently started a project to explore possibilities with AI in procurement.

All respondents in the study discussed the perceived benefits and values of using AI applications, as well as the challenges involved in their implementation. The specific benefits and challenges mentioned varied depending on the size of the agency, purchase volume and area of responsibility.

Several benefits and value contributions were highlighted, including improved operational capabilities and increased effectiveness in the procurement process. Time-saving benefits were particularly emphasized, as AI could free up time for more strategic tasks. In addition, respondents noted that AI applications could contribute to market development, innovation and environmental and social sustainability. These potential benefits were seen in all three major phases of the public procurement process.

Benefits and public value with artificial intelligence in public procurement

Examining the three broad phases of the public procurement process – preparation, execution and contract management – AI was considered by the interviewees as suitable and valuable in all three phases. In the preparation phase, respondents mentioned the potential use of AI in supply market research and analysis. They also pointed out that AI technology could be employed to discover new and more suppliers, as well as to gather information on suppliers' performance, including sustainability performance:

> It would be incredibly intriguing if we could find solutions in this area. We have a strong interest in discovering new suppliers, that is certain. We could envision working towards opening new markets, finding market data, and utilizing it in different ways as well. (Organization 10)

> We have the potential to enable a completely different analysis of the environment and market through AI technology. (Organization 8)

During the execution phase, two of the organizations included in the main study are utilizing AI applications. The organization that provides procurement business support to public

organizations, such as agencies and municipalities, is using a contractual tool that ensures quality based on best practices and legislation:

> It is a contractual tool that guarantees quality through best practices and legislation. I believe it's a great addition. With this tool, I can ensure that nothing is overlooked, focus on the most important aspects, and have a clear overview of the structure. The tool works effectively for consultancy agreements and confidentiality agreements. After submitting my text, it only takes a few minutes for the tool to highlight any deviations from the standard. (Organization 3)

Six other respondents also mentioned this use of AI as an area where it can contribute value. The second use of AI in public procurement, which can be categorized as part of the execution phase, is the AI solution used by one of the agencies which is viewed as part of the operative procurement work finding the right supplier and determine appropriate payment to supplier:

> We have a service for job seekers that started two years ago. This service involves 160 suppliers and 3,000 contracts. Many suppliers have multiple agreements. At the agency, we use a tool developed specifically for this service, which is based on AI. There are three different compensation levels based on the individual's distance from the labor market. Using AI, we can profile each participant and determine their corresponding level. Although the final decision on supplier and cost is made by an individual, I have an AI tool that compares the person's profile with our history of individuals with similar profiles and calculates the time it took for them to find a job. This solution helps us assess how far each individual is from the labor market and determines the appropriate payment to the supplier. (Organization 5)

This AI solution contributes to several values. First, it allows for an objective assessment of job seekers, eliminating the risk of discrimination or bias. Second, it facilitates a better match between job seekers and suitable suppliers. Finally, it provides better control over the amount of time spent on job seekers annually and helps to determine appropriate payments to suppliers.

In the final phase of contract management, respondents from six governmental agencies mentioned the potential use of AI for follow-up and evaluation. This included monitoring prices, costs and whether suppliers adhered to the environmental and social demands stated in the contracts. Few public agencies followed up if suppliers delivered according to agreements on products, services and costs, but also regarding environmental and social sustainability (including fraud and criminal activities among suppliers). The largest public procurement organization (in spend) in this study explicitly said that public procurement should contribute to developing supplier markets to become more innovative and sustainable. It is thus important not only to include innovation and sustainability in the preparation and execution phase. It is also of importance to follow up suppliers; do they deliver what has been agreed? Here AI, according to the respondents, might play a role and could contribute value.

Implementation challenges of artificial intelligence in public procurement

The implementation challenges of AI in public procurement were discussed by the respondents, who brought up various issues. These challenges included a lack of resources and low levels of knowledge in both AI and procurement. Inadequate data management and limited system support were also identified as challenges. Additionally, the lack of understanding and support from management was mentioned. Compliance with laws and regulations, as well as the high level of IT security required, posed obstacles. There were also problems with using external IT suppliers and a lack of knowledge about the market for AI solutions. Working in silos was another issue raised.

All respondents agreed that the lack of resources was a major challenge for increasing the overall maturity of procurement as well as for implementing AI initiatives. The respondents also highlighted the lack of support from management and the organization as a hindrance.

Consideration of laws and regulations, as well as the demand for high IT security and difficulties with external IT suppliers were common concerns among the respondents. The challenge of meeting high IT security requirements was specifically mentioned by five out of the ten organizations, particularly those involved in security and the management of personal data. In the workshop, this challenge was extensively discussed and further emphasized as a significant obstacle, especially for those agencies within security. Agencies also noted that they were not allowed to share information with each other due to legal restrictions, which hindered the use of AI in the preparation phase and management of contracts phase.

Ensuring suppliers have a good track record in environmental and social sustainability was deemed important, particularly to avoid potential criminal suppliers during the preparation phase. The importance of monitoring suppliers to ensure compliance with contractual obligations was also emphasized:

> The regulations have not been linked together so that information from the various agencies in Sweden can be accessed in order to be able to exclude suppliers who have committed a crime or have not paid taxes and fees. There should be a digital solution with real-time information about suppliers, one that aims to stop criminal suppliers and stop the money going to criminal activities. (National Agency for Public Procurement, General Manager, from pre-study).

To achieve this solution, a database must be created and managed over time, which should be fed with data from all agencies and other data sources. To accomplish this, it is necessary for

Table 2. Summary of empirical results

Procurement phase	AI application purpose/solution	Benefits/Value	Challenges
Preparation	Supply market research and analysis Tendering and supplier selection	Knowledge about, reach and cover new supplier markets Detect fraud and criminal behavior among potential suppliers Detect environmental and social behavior among potential suppliers	The consideration of laws and regulations, high level of IT security demand, problems with using external suppliers of IT solutions, lack of knowledge of the market (suppliers of AI solutions), working in silos
Execution	Contractual tool that can ensure quality based on best practice and legislation	Better operative capabilities Increase in procurement process effectiveness Time saving benefits	Lack of knowledge of the market (suppliers of AI solutions) Low level of knowledge about AI and benefits
Contract management	Follow-up and evaluation Monitoring contracts and suppliers	Secure that suppliers deliver what is agreed Secure sustainable suppliers Detect fraud and criminal behavior among suppliers Increased transparency	Lack of resources, the consideration of laws and regulations, high level of IT security demand, problems with using external suppliers of IT solutions, lack of knowledge of the market (suppliers of AI solutions) and working in silos

Source: Authors' own creation

agencies to collaborate instead of working independently. Empirical examples of benefits, values and challenges in different procurement phases are presented in Table 2 below.

Analysis of findings

One major observation is that of the governmental agencies included in the study only one of them had implemented an AI solution. This indicates a low level of AI maturity in terms of actual implementation and usage. A comparison with other European governmental agencies revealed similar findings. For instance, a report titled "Unlocking Public Sector AI" (2020) from the UK highlighted that "The low level of AI maturity within public procurement in Swedish governmental agencies is accompanied by a noticeable interest and recognition of what AI can offer". However, there are deficiencies in terms of AI preparedness, such as data management, AI technologies and internal knowledge and capabilities.

Perceived benefits and effects on value of artificial intelligence in public procurement

The perceived benefits of AI in public procurement were investigated through an abductive approach, based on the interviewees' perceptions and theory (Dubois and Gadde, 2002). The respondents faced time constraints in finding new suppliers, as there were few suppliers responding to tenders. Often, the suppliers were the same, lacking new innovative and sustainable ideas that could contribute to increased public value for citizens. The benefits of AI in terms of labor saving, cost effectiveness and time saving were primarily associated with reducing labor costs by automating tedious and time-consuming tasks. AI was also seen to have various benefits in information processing, which would require a stronger focus on data management strategies, including both internal and external data sources. In the context of public procurement, automation refers to AI systems replacing human work, while less attention has been given to AI for augmentation, which aims to enhance human intelligence in decision-making processes.

The perceived affordances/potentials of AI (Keller *et al.*, 2019) in the various phases of public procurement were associated with the preparation, execution and contract management phases (see Table 2). Different governmental agencies had varying degrees of association between AI and these phases. Just like in private business procurement, procurement managers in the government believed that AI technology could support the preparation phase by providing knowledge about new supplier markets, including those focused on innovation and sustainability. This included the use of AI for general supply market research, analysis and procurement strategy development. Some mentioned the potential use of AI in the execution phase, specifically in selecting suppliers. The contract management phase, which involves supplier and contract follow-up and evaluation, was frequently mentioned as an area where AI could be utilized. Among the governmental agencies, monitoring contracts and ensuring adherence to social sustainability requirements in the contract management phase were seen as important areas where AI could be applied. Additionally, AI was seen as a useful tool for controlling fraud and criminal activities.

When it comes to the public and citizen *value* (Malacina *et al.*, 2022) of incorporating AI into public governmental procurement, the use of AI in procurement was described to improve sustainability by monitoring green public procurement, reducing carbon footprints and ensuring social sustainability throughout the procurement process. Additionally, AI could according to public interviewees be employed to enhance compliance efforts, detect and prevent fraud and monitor labor policies in supply chains more efficiently.

In terms of market development and performance, larger governmental agencies with significant purchasing power see AI as a potential tool for driving supply market competition, growth and development. AI can shorten procurement cycles, influence supplier markets and

provide market insights, making it easier for small and medium-sized enterprises (SMEs) to access supplier markets and foster competition.

Overall, increased efficiency and improved operative capabilities throughout the procurement process were frequently cited as the most significant advantages of AI. Implementing AI was expected to optimize time management and utilize existing internal resources more effectively. AI could automate routine tasks such as document processing, data entry and data analyses, leading to quicker and more efficient risk assessments, supplier reporting, compliance monitoring and more. AI was also expected to enhance the effectiveness of the public procurement process by ensuring supplier compliance, reducing errors, controlling corruption and improving public service quality and availability through enhanced supplier selection, evaluation, compliance and quality standards monitoring, as well as performance monitoring and feedback.

In line with Enholm *et al.* (2021) research, interviewees in public procurement described the anticipated outcomes of AI implementation in governmental procurement in terms of both immediate effects on procurement processes and broader societal impacts. Potential immediate effects included improved procurement efficiency, productivity and decision-making quality. Secondary societal effects mentioned included long-term improvements in public services, enhanced services, citizen satisfaction and effects on supplier management and supplier markets. Powerful governmental agencies responsible for large-scale infrastructure projects emphasized many of these indirect, secondary societal effects resulting from increased efficiency and effectiveness. Overall, incorporating AI into public and governmental procurement was believed to have the potential to generate significant benefits for the public, citizens and the procurement process as a whole. Hence, both first-order and second order effects (Enholm *et al.*, 2021) were mentioned as potential benefits which, when and if implemented, could have effects on the delivery of different forms of public value public.

Perceived artificial intelligence implementation challenges

As regards implementation challenges, one organizational challenge (Enholm *et al.*, 2021) mentioned was the lack of top management support within government agencies. This was described as often being combined with an old, established silo-based organizational structure and a culture that does not promote a progressive view on digitalization, AI and new technologies in procurement. Some agencies also mentioned a lack of knowledge and internal capabilities, including knowledge about potential external technology partners and access to external AI capabilities and cooperation with other agencies.

Regarding the implementation process, the implementation of AI had in most cases not started yet, so there were limited experiences with concrete implementation process challenges. However, there were concerns that there might be resistance and inertia in the procurement process due to lack of AI capabilities, technical expertise and clear responsibilities and accountability.

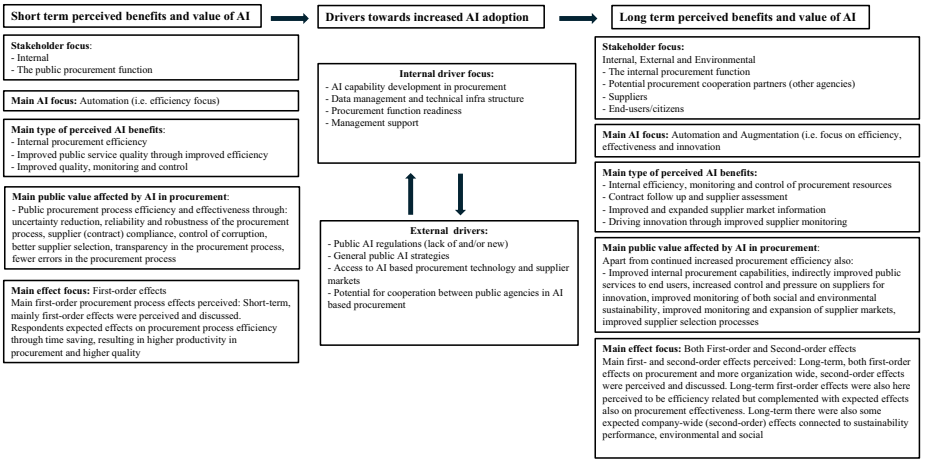
In terms of technology challenges (ibid), some public procurement managers expressed concerns about data, as identified by Merhi (2023) and Nowicki (2020). Issues such as low data quality, insufficient quantity of data, data security and confidentiality and other data governance challenges were mentioned.

Legal concerns surrounding the implementation of AI in government agencies were also highlighted. Regulations, security risks, legal constraints and lack of technology standards were mentioned as major challenges. The identification and selection of vendors for AI solutions were also mentioned, although not as a central issue, but more as a need for general information about potential vendors and their AI-based value propositions.

Short- and long-term effects of artificial intelligence implementation

Looking at the empirical data in relation to the research questions and the conceptual framework, there was *a strong link between the perceived benefits, the value outcomes and the implementation challenges*. *First*, from a temporal perspective the views of the governmental procurement managers were often presented either in a short-term or a long-term perspective, which both were connected to a more internal, procurement related orientation (short-term) or a wider, governmental agency effectiveness or societal value focused orientation (long-term). In other words, the views on AI in governmental procurement were positioned in either a short-term or a long-term perspective. *Second*, the perceived benefits and value created by AI shifted in the short-term case from a focus on the internal value of the public procurement buyer, to a wider stakeholder value perspective in the long-term case to include comments also for suppliers and to end users, i.e. citizens. *Third*, while the short-term perspective was dominated by (internal) efficiency benefits of AI in procurement, the long-term view expanded this to include comments also on effectiveness, public service quality, supply market efficiency and various societal benefits. Interviewees also commented on crucial factors or drivers for public procurement to go from short-term interest and tentative test of AI to a more elaborate and stable use. Ideas and comments of such drivers toward stabilized AI use included both internal, often capability related factors, but also external, both institutional and supplier market drivers (Figure 1).

The interview study also highlighted that some aspects are emphasized more when AI is implemented in public procurement. It was frequently mentioned that public procurement is subject to legal frameworks and stringent regulations for transparency, accountability and fairness. As a result, according to the procurement managers, AI systems need to be adapted and carefully controlled to comply with these requirements. The use of AI in public procurement necessitates transparency to maintain accountability, which can impact the tendering processes and the trust among private suppliers as well as the trust of the public, including citizens and recipients of public services, according to the interviewees.



Source: Authors' own work

Figure 1. Perceptions of the shift between short-term and long-term adoption and use of AI in public procurement

In comparison to private procurement, public procurement was perceived as somewhat more constrained in terms of using external AI vendors, such as external cloud services. Additionally, the legacy systems within governmental agencies, including fragmented data and data silos, are seen as creating certain challenges in public procurement when implementing AI.

Discussion

Overall, few Swedish government teams had procured AI solutions in the context of AI-based procurement. This is in contrast to the "Government AI Readiness Index 2021" (Oxford Insights, 2021), which positions Sweden as a global leader in AI maturity and concludes that "the Nordic nations tend to score well because of their governments' high internal capabilities, implying that they will successfully manage AI projects." This aligns with previous studies on AI and private company procurement (Guida *et al.*, 2023; Meyer and Henke, 2023; Allal-Chérif *et al.*, 2021). Notably, studies focusing on private procurement (Guida *et al.*, 2023; Meyer and Henke, 2023) also struggle to find empirical data demonstrating the actual usage of AI tools or technology in procurement work.

Benefits and value

As regards AI benefits and value, compared to previous research on the use of AI in both public and private procurement, this study confirms many of the perceived benefits and potential outcomes and effects (Sanchez-Graells, 2024a; Cui *et al.*, 2022; Buchholz and Grabbe, 2022; Nowicki, 2020). For example, it supports the hopes for improved operational efficiency in different steps of the tendering and procurement process (Cui *et al.*, 2022; Buchholz and Grabbe, 2022). Internal procurement efficiency, leading to improved public service quality and improved quality monitoring and control were frequently mentioned. These perceptions can be compared to the ten benefits of AI in public management identified by Wirtz *et al.* (2021). Recurrently, the interviewees mentioned the benefits of broader accessibility and availability, particularly in terms of reaching and covering new supplier markets. Although AI has been recognized as a technology that supports supplier selection in private procurement (Guida *et al.*, 2023), in this study, procurement professionals saw AI as a tool for obtaining knowledge on how to reach and cover new supplier markets. Similarly to the benefits identified by Wirtz *et al.* (2021), the interviewees emphasized the importance of monitoring and transparency in public procurement, particularly in relation to sustainability and detecting criminal behavior in supply chains. These findings align with previous studies on AI in private procurement, which highlight procurement risk mitigation as a key benefit of AI (Guida *et al.*, 2023; Brintrup *et al.*, 2020). The perceived benefits of AI in public procurement revolve around increased efficiency and automation. Enholm *et al.* (2021) propose that AI applications can be broadly categorized as either AI for automation or AI for augmentation. Some of the benefits are directly related to the general values generated by public procurement, as described by Malacina *et al.* (2022). The study shows (see Figure 1) that like in private procurement, automation is the initial short-term focus, while augmentation through AI is perceived as bringing benefits to public procurement in a long-term perspective. Sustainability monitoring (environmental, social and economic) with the help of AI in public procurement leading to both improved and to new public value (cf Malacina *et al.*, 2022) was frequently mentioned.

In line with Enholm *et al.* (2021) research, interviewees described the anticipated outcomes of AI implementation in governmental public procurement in terms of both immediate effects on procurement processes and broader societal impacts. Potential *immediate effects* include improved procurement efficiency, productivity and decision-making quality. *Secondary societal effects* mentioned include long-term improvements in

public services, enhanced services, citizen satisfaction and effects on supplier management and supplier markets.

Implementation challenges

According to Merhi's (2023) study on AI implementation challenges, there are four different types of challenges that public procurement managers can face: *organization*, *process*, *technology and environment*. This can be compared with Wirtz *et al.* (2019) who presented an AI-challenges model summing up four major forms of challenges of AI in public management in general: AI technology implementation, AI law and regulations, AI ethics and AI society. Overall, the public procurement functions in the study were mainly concerned with the first category which includes challenges concerning, e.g. system/data quality and integration, specialization and expertise, AI safety and financial feasibility of the new procurement systems. Public procurement laws and regulation, ethics etc. were perceived as given factors which the procurement functions still needed to adapt, if and when AI was adopted. Compared to studies on AI and private procurement, challenges found in this study are fairly similar. In the study by Guida *et al.* (2023), a main obstacle was the availability and quality of data to be processed, which is a main obstacle or challenge found also in this study. IT safety is also found to be a main challenge in this study. High IT safety demands make it difficult, if not impossible, for some of the agencies to use external AI suppliers. IT security has not been discussed to large extent in studies on AI and private procurement (Guida *et al.*, 2023; Meyer and Henke, 2023). Meyer and Henke (2023) raise data protection and data security aspects when discussing requirements for AI in private procurement, they do not however raise IT security as requirement.

Looking at organization challenges, the lack of top management support within the agencies for procurement as such was found to be a challenge. Meyer and Henke (2023) bring up top management support as a requirement and claim that top management should support the implementation of AI in purchasing and supply management. Lack of skills and resources was brought up by the respondents as a challenge. This is in line with Guida *et al.* (2023), and also Meyer and Henke (2023) argue for competence development within the purchasing department as a requirement.

In summary, in comparison to previous research, this study revealed that the perceived benefits of AI were mainly focused on improved operative capabilities, enhanced process effectiveness and the potential for AI to improve sustainability monitoring. Specifically, the use of AI in the preparation and contract management phases showed potential for generating social sustainability value by detecting criminal activities and corruption among suppliers. This could lead to the avoidance or termination of contracts with such suppliers, thereby contributing to social sustainability.

Conclusions and theoretical and managerial implications

Conclusions

The aim of this paper is to contribute to knowledge regarding the perceived benefits of AI in public procurement and the challenges associated with its implementation, as well as how AI can contribute to the creation of value in public procurement. The perceived benefits of AI primarily revolved around improved operative capabilities, certain potential for process effectiveness and the potential for improved monitoring through AI. Public procurement managers identified three categories of challenges: organizational challenges resulting from a silo-based organizational structure, process challenges related to uncertain responsibilities and accountability and technological challenges tied to data management issues. Agencies within the realm of security and those handling large amounts of personal data highlighted IT

safety as a major challenge. On the other hand, larger agencies responsible for large-scale building construction projects emphasized the higher values and societal effects of increased efficiency and effectiveness, while smaller agencies tended to prioritize the efficiency of the procurement process. Findings also show that the perceived benefits and value created by AI can be viewed from a short-term perspective to a long-term perspective. From a focus on the internal value of the public procurement buyer, to a wider stakeholder value perspective in the long-term case to include comments also for suppliers and to end users, i.e. citizens.

Theoretical contributions and implications

The study contributes to research on public procurement by providing knowledge on the benefits, values and challenges associated with the use and potential use of AI applications in public procurement. Previous research has indicated that there are several opportunities with using AI technologies in public procurement, but also challenges (Sanchez-Graells, 2024a; Abrahamsson and Behr, 2022; Nowicki, 2020). This study contributes with empirical insights and knowledge regarding the perceived benefits of AI in public procurement and the challenges associated with its implementation from the public procurement professional perspective. The study also contributes to knowledge on how AI can contribute to the creation of public value through public procurement, thus adding knowledge to previous studies that mainly have focused on legal and ethical aspects (Sanchez-Graells, 2024a; Abrahamsson and Behr, 2022). Another contribution to literature concerns perceptions of the shift between short-term and long-term adoption and use of AI in public procurement. Previous studies on AI in public procurement and AI in private procurement has not clearly separated these two perspectives (Guida *et al.*, 2023; Meyer and Henke, 2023; Allal-Chérif *et al.*, 2021; Nowicki, 2020). We also show the usefulness of previous studies on AI in public management in general.

(Wirtz *et al.*, 2021; Wirtz and Muller, 2019; Wirtz *et al.*, 2019). Therefore, this study also contributes to the existing public procurement literature by presenting a conceptual framework that encompasses the key dimensions: perceived benefits (drivers, reasons, motivations and goals), implementation challenges (problems, difficulties) and outcomes on public value (effects, impacts, changes) when AI is implemented.

Managerial implications

The conceptual framework and its four central dimensions can serve as a starting point for public procurement organizations in the process of planning for the implementation of AI in their procurement processes. First, what is the main motive for implementing AI in procurement operations? The framework and empirical study suggest several different types of reasons for using AI in procurement that go beyond simple efficiency gains. A practical implication is that a thorough analysis of the most important reasons why AI, what AI can achieve and what problems it can solve is crucial as it will also impact the other general dimensions. Second, the framework can also assist practitioners in anticipating potential challenges that may arise during the implementation of AI. The literature and empirical study have shown that some challenges are common in procurement, while others are specific to public procurement. Third, a comprehensive analysis of the motives for implementing AI will also indicate which specific procurement processes should initiate the change process. This is an important practical implication as it will gradually affect other parts of the procurement processes, operations and internal as well as external organizations. Fourth, the framework also provides practitioners with a foundation for understanding the potential consequences of AI implementation in public procurement, both in terms of direct procurement activities and organization, and in terms of broader impacts on public administration and public value creation.

Limitations and future studies

Although the majority of agencies had plans and insights into AI, more empirical studies are needed investigating concrete implementations and effects. Only two of the agencies were using AI technologies in their procurement work, therefore further knowledge is required about the practical value of AI in public procurement. The current research is primarily based on the accounts of procurement managers regarding perceived benefits, but it is necessary to gather knowledge through actual experiences of using AI in public procurement. This includes understanding the contribution of AI to various forms of public services. Additionally, a deeper understanding of the challenges in implementing AI is needed. Conducting a longitudinal case study that tracks the implementation of an AI tool in an agency over an extended period of time would provide valuable insights into the implementation process.

Other recommendations for future studies include focusing on specific challenges identified in this research and conducting more in-depth investigations. For instance, one suggested area of focus is the challenge of IT security in using AI for public procurement. A comparative study could examine the IT security requirements for public procurement in different countries and explore potential methods to meet these requirements while also utilizing external AI suppliers to support procurement work. Furthermore, there is a need to examine the competence in AI, both internally and externally, in the context of public procurement. To advance knowledge in this area, it is suggested that future research investigates the collaboration between various stakeholders, both internal and external, to ensure successful implementation and utilization of AI in public procurement. As far as the authors know, no such studies have been conducted thus far.

References

Abrahamsson, P. and Behr, T. (2022), "AI governance and ethics in public procurement: bridging the gap between theory and practice", *IEEE 28th International Conference on Engineering, Technology and Innovation and 31st International Association for Management of Technology (IAMOT) Joint Conference*, June, pp. 19-23, 2022, *Nancy.*

Allal-Chérif, O.V., Simón-Moya, P. and Ballester, J.A.C.C. (2021), "Intelligent purchasing: how artificial intelligence can redefine the purchasing function", *Journal of Business Research*, Vol. 124, pp. 69-76, doi: 10.1016/j.jbusres.2020.11.050.

Braun, V. and Clarke, V. (2006), "Using thematic analysis in psychology", *Qualitative Research in Psychology*, Vol. 3 No. 2, pp. 77-101.

Brintrup, A., Pak, J., Ratiney, D., Pearce, T., Wichmann, P., Woodall, P. and McFarlane, D. (2020), "Supply chain data analytics for predicting supplier disruptions: a case study in complex asset manufacturing", *International Journal of Production Research*, Vol. 58 No. 11, pp. 3330-3341.

Buchholz, W., Pak, J. and Grabbe, F. (2022), "Application of digital technologies in procurement: potential analysis based on the three-layer model", in Bode, C., Bogaschewsky, R., Eßig, M., Lasch, R. and Stölzle, W. (Eds), *Supply Management Research. Advanced Studies in Supply Management*, Springer Gabler, Wiesbaden, doi: 10.1007/978-3-658-38350-3_11.

Constant, F.R.E., Amarani, M., Berre, P. and Hameed, M.S.S. (2022), "Perceived benefits of adopting artificial intelligence technologies in purchasing processes", *IPSERA conference, Jonkoping.*

Cui, R.,Li, S. and Zhang, M. (2022), "AI and procurement", *Manufacturing and Service Operations Management*, Vol. 24 No. 2, pp. 691-706.

Dubois, A. and Gadde, L.-E. (2002), "Systematic combining: an abductive approach to case research", *Journal of Business Research*, Vol. 55 No. 7, pp. 553-560.

Enholm, I.M.,Papagiannidis, P., Mikalef, J. and Krogstie, E. (2021), "Artificial intelligence and business value: a literature review", *Information Systems Frontiers*, Vol. 24 No. 5, pp. 1709-1734.

Guida, M.,Caniato, A., Moretto, S. and Ronchi, F. (2023), "The role of artificial intelligence in the procurement process: state of the art and research agenda", *Journal of Purchasing and Supply Management*, Vol. 29 No. 2.

Jahani, N.,Sepehri, H.R., Vandchali, E.B. and Tirkolaee, A. (2021), "Application of industry 4.0 in the procurement processes of supply chains: a systematic literature review", *Sustainability*, Vol. 13 No. 14, pp. 1-25.

Kaplan, A. and Haenlein, M. (2020), "Rulers of the world, unite! the challenges and opportunities of artificial intelligence", *Business Horizons*, Vol. 63 No. 1, pp. 37-50.

Keller, R., Stohr, A., Fridgen, G., Lockl, J. and Rieger, A. (2019), "Affordance-experimentation-actualization theory in artificial intelligence research - A predictive maintenance story".

Kiefer, D., Ulmer, A. and Dinther, C.V. (2019), *Application of Artificial Intelligence to Optimize Forecasting Capability in Procurement*, Wissenschaftliche Vertiefungskonferenz-Tagungsband 2019, pp. 69-80.

Knutsson, H. and Thomasson, A. (2014), "Innovation in the public procurement process: a study of the creation of innovation-friendly public procurement", *Public Management Review*, Vol. 16 No. 2, pp. 242-255.

Kosmol, T., Reimann, F. and Kaufmann, L. (2019), "You'll never walk alone: Why we need a supply chain practice view on digital procurement", *Journal of Purchasing and Supply Management*, Vol. 25 No. 4, p. 100553.

Lu, Y. (2019), "Artificial intelligence: a survey on evolution, models, applications and future trends", *Journal of Management Analytics*, Vol. 6 No. 1, pp. 1-29.

Madan, R. and Ashok, M. (2023), "AI adoption and diffusion in public administration: a systematic literature review and future research agenda", *Government Information Quarterly*, Vol. 40 No. 1, p. 101774.

Malacina, I.,Jääskeläinen, K., Lintukangas, J. and Heikkilä, A. (2022), "Capturing the value creation in public procurement: a practice-based view", *Journal of Purchasing and Supply Management*, Vol. 28 No. 2.

Markus, M. and Silver, M. (2008), "A foundation for the study of IT effects: a new look at DeSanctis and Poole's concepts of structural features and spirit", *Journal of the Association for Information Systems*, Vol. 9 No. 10, doi: 10.17705/1jais.00176.

Merhi, M.I. (2023), "An evaluation of the critical success factors impacting artificial intelligence", *International Journal of Information Management*, Vol. 69, p. 102545.

Meyer, M. and Henke, M. (2023), "Developing design principles for the implementation of AI in PSM: an investigation with experts", *Journal of Purchasing and Supply Management*, Vol. 29 No. 3, p. 100846.

Niyazbekova, S.U., Kurmankulova, R.Z., Anzorova, S.P., Goigova, M.G. and Yessymkhanova, Z.K. (2021), "Digital transformation of government procurement on the level of state governance", in Popkova, E.G., Ostrovskaya, V.N., Bogoviz, A.V. (Eds), *Socio-Economic Systems: Paradigms for the Future. Studies in Systems, Decision and Control*, Springer, Cham, Vol. 314, doi: 10.1007/978-3-030-56433-9_69.

Nowicki, P. (2020), "Deus ex machina?", *European Procurement and Public Private Partnership Law Review*, Vol. 15 No. 1, pp. 53-60.

Oxford Insights (2021), "Government AI readiness index 2021", available at: www.oxfordinsights.com/government-ai-readiness-index2021 (accessed 29 Aug 2023).

Rejeb, A.,Sűle, J. and Keogh, E. (2018), "Exploring new technologies in procurement", *Transport and Logistics: The International Journal*, Vol. 18 No. 45.

Sanchez-Graells, A. (2024a), *Digital Technologies and Public Procurement: Gatekeeping and Experimentation in Digital Public Governance*, Oxford University Press, Oxford.

Sanchez-Graells, A. (2024b), "Public procurement of artificial intelligence: recent developments and remaining challenges in EU law", LTZ (Legal Tech Journal) 2/2024, available at: https://ssrn.com/abstract=4706400

Saunders, M.N.K., Lewis, P. and Thornhill, A. (2023), *Research Methods for Business Students*, 9th edition, Pearson.

Schulze-Horn, I.,Hueren, P., Scheffler, H. and Schiele, S. (2020), "Artificial intelligence in purchasing: facilitating mechanism design-based negotiations", *Applied Artificial Intelligence*, Vol. 34 No. 8, pp. 618-642.

Seale, C., Gabo, G., Gubrium, J. and Silverman, D. (2004), *Qualitative Research Practice*, Sage, London.

Siciliani, L., Taccardi, V., Basile, P., Di Ciano, M. and Lops, P. (2023), "AI-based decision support system for public procurement", *Information Systems*, Vol. 119, p. 102284.

Spreitzenbarth, J.M., Bode, C. and Stuckenschmidt, H. (2024), "Artificial intelligence and machine learning in purchasing and supply management: a mixed-methods review of the state-of-the-art in literature and practice", *Journal of Purchasing and Supply Management*, Vol. 30 No. 1, p. 100896.

Spreitzenbarth, J.,Stuckenschmidt, C. and Bode, H. (2021), "Methods of artificial intelligence in procurement: a conceptual literature review", *IPSERA Conference, Online Conference*.

Strong, D.M., Johnson, S.A., Tulu, B., Trudel, J., Volkoff, O., Pelletier, L.R., Bar-On, I. and Garber, L. (2014), "A theory of organization-EHR affordance actualization", *Journal of the Association for Information Systems*, Vol. 15 No. 2, pp. 53-85.

Thompson, J. (2022), "A guide to abductive thematic analysis", *The Qualitative Report*, Vol. 27 No. 5, pp. 1410-1421.

Wirtz, B.W., Langer, P.F. and Fenner, C. (2021), "Artificial intelligence in the public sector - a research agenda", *International Journal of Public Administration*, Vol. 44 No. 13, pp. 1103-1128, doi: 10.1080/01900692.2021.1947319.

Wirtz, B.W. and Muller, W.M. (2019), "An integrated artificial intelligence framework for public management", *Public Management Review*, Vol. 21 No. 7, pp. 1076-1100, doi: 10.1080/14719037.2018.1549268.

Wirtz, B.W., Weyerer, J.C. and Geyer, C. (2019), "Artificial intelligence and the public sector – applications and challenges", *International Journal of Public Administration*, Vol. 42 No. 7, pp. 596-615.

Yin, R. (2018), *Case Study Research: Design and Methods*, 6th Edition, Sage publications.

Zhang, C. and Lu, Y. (2021), "Study on artificial intelligence: the state of the art and future prospects", *Journal of Industrial Information Integration*, Vol. 23, pp. 1-9.

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